

Quality-Stratified Multimodal Fusion for Skin Lesion Malignancy Classification on MCR-SL

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Why This Problem?

- Deep learning has reached dermatologist-level accuracy on **image-only** skin lesion classification (Esteva et al., 2017; HAM10000).
- But a real dermatologist never diagnoses from an image alone — age, sex, lesion site, referral history all matter.
- A smaller line of work fuses images with **patient metadata** (e.g. MetaBlock on PAD-UFES-20) and shows it helps.
- **Missing piece:** none of these datasets record how *good* the diagnostic image itself was.
 - Real clinical images vary hugely in quality (lighting, focus, hair, blur).
 - No prior benchmark could ask: *do errors concentrate on poor-quality images?*

The Opportunity: MCR-SL

A brand-new dataset (Castro-Fernandez et al., *Data*, 2025) closes this gap:

- Every diagnostic image has an expert-assigned **1–10 quality rating**.
- Rich structured metadata per lesion/subject (22 fields).
- Partial histopathology confirmation (ground-truth beyond panel opinion).

Key fact

No prior published benchmark exists on MCR-SL. We are first.

This lets us ask a question no earlier skin-lesion benchmark could:

Does model performance degrade on lower-quality images, and can we train the model to be more robust to that?

MCR-SL Dataset at a Glance

Property	Value
Subjects	60
Lesions (total / usable for binary task)	240 / 234
Malignant / non-malignant / undetermined	42 / 192 / 6
Clinical images (usable)	779
Dermoscopic images (usable)	1352
Histopathology-confirmed lesions	28
Metadata fields used (categorical / numeric)	16 / 4
Lesions with full expert quality coverage	238
Unified diagnosis classes	9

“Usable” = has a lesion record match and an existing image file on disk.

Task Definition

Primary task: Binary classification

Malignant vs. non-malignant, per lesion.

Ground truth: `unified_diagnosis` field — histopathology-confirmed where available, expert-panel consensus otherwise.

Secondary / exploratory task: 9-class diagnosis

NEV, SK, BCC, AK, ATY, MEL, SCC, ANG, DF.

Caveat: several classes have <10 lesions (MEL=8, SCC=5, ANG=4, DF=2) — reported as exploratory only, never as robust per-class statistics.

What This Work Does

- ① First published benchmark on MCR-SL: image-only vs. late-fusion vs. **channel-gated fusion** (our main method), under a rigorous subject-disjoint 5-fold protocol.
- ② Four **robustness analyses** enabled uniquely by MCR-SL's quality ratings and histopathology subset:
 - Quality-stratified performance
 - Quality-aware training (auxiliary objective)
 - Histopathology-confirmed vs. panel-consensus comparison
 - Metadata-importance audit vs. the dataset paper's own statistics
- ③ A sweep of 7 further interventions (11 configs total: SAM, focal loss, preprocessing, contrastive loss, TTA, multi-image averaging, and combinations) — honestly reporting what helped and what didn't.
- ④ Best result: **0.822 balanced accuracy**, close to the best verified published binary result on a comparable dataset (0.836) with far less training data.

Architecture — Two Encoders

Image Encoder

- EfficientNet-B0, ImageNet-pretrained
- Input: dermoscopic image, 224×224
- Output: 1280-d pooled vector *and* the last conv feature map ($1280 \times 7 \times 7$)

Metadata Encoder

- 16 categorical fields $\rightarrow$ learned embeddings (with explicit “unknown” category — *no imputation*)
- 4 numeric fields (age, height, weight, diameter) $\rightarrow$ z-scored using *train-fold-only* statistics + missingness bit
- Concat $\rightarrow$ 2-layer MLP $\rightarrow$ 128-d metadata vector

Both encoders feed into one of two fusion strategies (next slide).

Architecture — Two Fusion Strategies

1. Late fusion (baseline)

concat(pooled image vector, metadata vector) $\rightarrow$ Linear+ReLU+Dropout (256-d) $\rightarrow$ classifier.
This is the standard MetaBlock-style approach from prior literature.

2. Channel-gated fusion (our main method)

metadata vector $\rightarrow$ Linear $\rightarrow$ Sigmoid $\rightarrow$ a 1280-d **gate** $\in [0, 1]^{1280}$
 $\rightarrow$ multiply elementwise against the image's conv feature map, channel-by-channel
 $\rightarrow$ global pool $\rightarrow$ same 256-d projection $\rightarrow$ classifier.

Idea: let patient metadata decide **which visual channels matter**, instead of just concatenating two vectors. Same mechanism as Squeeze-and-Excitation, but the gating signal comes from metadata, not from the image itself.

Architecture — Three Prediction Heads

All three heads sit on top of the **same** 256-d fused vector; only their outputs and losses differ.

1. Binary head (main task, always on)

256-d $\rightarrow$ Linear $\rightarrow$ 1 logit $\rightarrow$ sigmoid $\rightarrow P(\text{malignant})$.

Loss: weighted BCE (or focal loss), class weights recomputed per training fold ($\sim 1:4.6$ malignant:non-malignant).

This is the only head whose output is reported as the paper's headline results.

2. Auxiliary 9-class head (exploratory)

256-d $\rightarrow$ Linear $\rightarrow$ 9 logits $\rightarrow$ softmax $\rightarrow$ unified diagnosis (NEV, SK, BCC, ...).

Loss: class-weighted CE, downweighted to $\sim 0.3\text{--}0.5\times$ the binary loss.

Reported only as an exploratory table — several classes have <10 lesions.

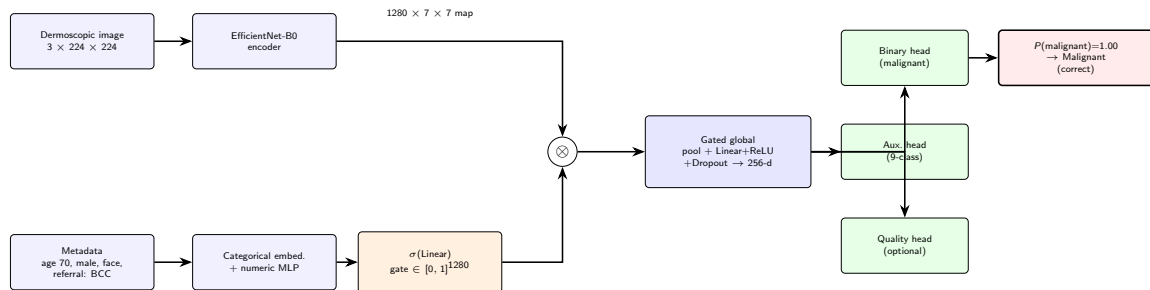
3. Quality-regression head (optional, quality-aware variant only)

256-d $\rightarrow$ Linear $\rightarrow$ 1 scalar $\rightarrow$ predicted mean expert quality rating (1–10).

Loss: MSE, weighted $0.1\text{--}0.2\times$.

Tests whether explicit quality-awareness improves robustness — it did not (see Robustness Analyses).^{10 / 27}

Full Pipeline (with a Real Example)



Example: lesion L0189, histopathology-confirmed basal cell carcinoma — one of the model's most confident correct predictions.

Training / Evaluation Protocol

Subject-disjoint stratified 5-fold cross-validation

- Split by `subject_id`, never by lesion — a subject can have multiple lesions, and images of the same lesion are near-duplicates.
- Folds balanced by greedily equalizing malignant-lesion counts per fold.
- For each rotation: 1 fold held out for **test**, 1 disjoint fold held out only for **checkpoint selection** (val), rest for training.
- **No hyperparameter tuning against final test-fold results.**

Losses

- Binary head: weighted BCE, class weights recomputed per training fold ($\sim 1:4.6$ malignant:non-malignant overall).
- Auxiliary 9-class head: class-weighted CE, downweighted ($\sim 0.3\text{--}0.5\times$), exploratory only.
- (Optional) Quality-regression head: low weight ($0.1\text{--}0.2$), predicts mean expert rating.

The 5 Folds, Visualized

60 subjects $\rightarrow$ 5 disjoint folds (balanced by malignant-lesion count). Every rotation uses a **different** test fold and a different, disjoint val fold — so all 5 folds eventually serve as the test fold exactly once.

Rotation	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
1	Test	Val	Train	Train	Train
2	Train	Test	Val	Train	Train
3	Train	Train	Test	Val	Train
4	Train	Train	Train	Test	Val
5	Val	Train	Train	Train	Test

- **Train** folds: model weights are updated here.
- **Val** fold: used only to pick the best training checkpoint (never influences weights).
- **Test** fold: touched exactly once, only for final reported metrics — never used for any decision.

Final numbers = mean $\pm$ std of the 5 **test**-fold results (one per rotation).

Evaluation Metrics — What Each One Means

Metric	Definition	Why it matters here
Accuracy	(correct predictions) / (all predictions)	Can be misleading: predicting “non-malignant” always still scores $\sim 82\%$
Balanced accuracy	average of sensitivity and specificity	Primary metric — corrects for the malignant class being a minority ($\sim 18\%$)
Macro-F1	F1 (precision/recall harmonic mean) averaged equally across both classes	Rewards doing well on <i>both</i> classes, not just the majority one
Sensitivity (recall)	(malignant caught) / (all actual malignant)	Missed malignancy = missed cancer diagnosis — the costliest error clinically
Specificity	(non-malignant cleared) / (all actual non-malignant)	Low specificity means unnecessary biopsies/patient anxiety
AUROC	probability the model ranks a random malignant lesion above a random non-malignant one	Threshold-independent measure of overall ranking quality

All reported as mean $\pm$ std across the 5 test folds, plus a confusion matrix aggregated across all folds.

Core Ablation: Image-Only vs. Late Fusion vs. Channel-Gated

Metric	Image-only	Late fusion	Channel-gated
Accuracy	0.813 ± 0.020	0.831 ± 0.052	0.833 ± 0.070
Balanced acc.	0.784 ± 0.073	0.768 ± 0.090	0.783 ± 0.085
Macro F1	0.733 ± 0.038	0.743 ± 0.051	0.757 ± 0.073
Sensitivity	0.697 ± 0.187	0.644 ± 0.191	0.672 ± 0.159
Specificity	0.872 ± 0.061	0.891 ± 0.021	0.895 ± 0.048
AUROC	0.895 ± 0.053	0.851 ± 0.080	0.881 ± 0.065

Channel-gated wins macro-F1, but does **not** dominate every metric.

Image-only actually has higher sensitivity and AUROC — reported honestly, not hidden.

Balanced accuracy is effectively tied between image-only and channel-gated (0.784 vs. 0.783).

Quality-Aware Variant — A Negative Result

Added an auxiliary head to predict image quality, hoping it would make the model more robust.

Metric	Channel-gated	+ Quality-aware
Balanced accuracy	0.783	0.740
Sensitivity	0.672	0.578
Specificity	0.895	0.902
AUROC	0.881	0.879

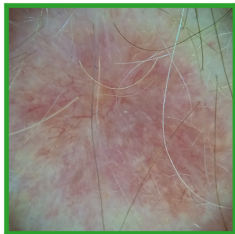
It **underperformed** on 5 of 6 metrics — became more conservative (small specificity gain, big sensitivity loss). Worse direction for a cancer-screening task. Reported honestly as a negative result.

Follow-Up Intervention Sweep (11 configurations)

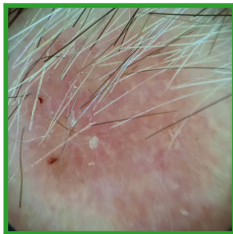
Configuration	Acc.	Bal. acc.	Sens.	Spec.	AUROC
Focal loss	0.858	0.809	0.694	0.923	0.877
Dermoscopy preprocessing	0.828	0.793	0.736	0.851	0.875
Contrastive auxiliary loss	0.812	0.770	0.669	0.872	0.873
Alt. optimizer (AdamW+cosine)	0.822	0.757	0.661	0.853	0.845
Test-time augmentation (TTA)	0.841	0.788	0.672	0.904	0.897
Multi-image averaging	0.854	0.813	0.719	0.907	0.906
Sharpness-aware min. (SAM)	0.858	0.810	0.694	0.927	0.904
SAM + multi-image	0.851	0.800	0.672	0.928	0.917
SAM + TTA	0.861	0.822	0.719	0.925	0.908
SAM + TTA + multi-image	0.856	0.810	0.694	0.925	0.915
Preprocess.+focal+SAM+TTA	0.818	0.759	0.622	0.895	0.893

Best: SAM + TTA → 0.822 balanced accuracy. Stacking everything together **hurt** rather than helped.

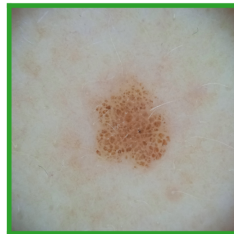
Example Predictions (Best Method)



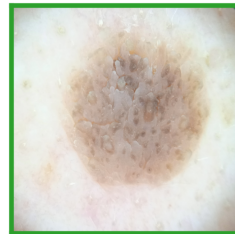
Age 70 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, $P=1.00$



Age 57 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, $P=1.00$



Age 50 / Female / Back
Referral: Voluntary sample
Ground truth: Non-malignant
Predicted: Non-malignant, $P=0.00$



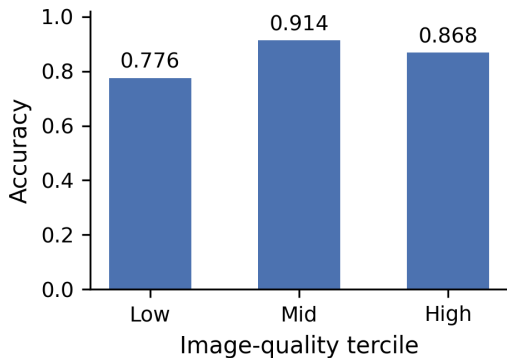
Age 64 / Male / Back
Referral: SK
Ground truth: Non-malignant
Predicted: Non-malignant, $P=0.00$

Highest-confidence *correct* predictions — illustrative only, not representative of overall performance (see full metrics above).

Why Robustness Analyses Are the Real Contribution

- Beating a prior number was never really possible — **no prior benchmark exists on MCR-SL.**
- What MCR-SL uniquely offers: per-image expert quality ratings + partial histopathology confirmation.
- So the paper's actual novelty is asking questions no earlier skin-lesion benchmark could ask:
 - ① Does performance drop on lower-quality images?
 - ② Can we train the model to be more robust to quality variation?
 - ③ Does performance differ on histopathology-confirmed vs. panel-consensus lesions?
 - ④ Does the model “look at” the same metadata fields that the dataset's own statistics flagged as significant?

1. Quality-Stratified Performance



Tercile	N	Acc.	Sens.
Low	85	0.776	0.667
Mid	93	0.914	0.733
High	53	0.868	0.500

Pattern is **not monotonic**. Spearman correlation (rating vs. error) = -0.093 , $p = 0.158$ — not significant at $N = 231$. No strong quality-performance relationship detected (neither confirmed nor ruled out).

2. Quality-Aware Training — Did It Help?

Tercile	N	Acc. (plain)	Acc. (quality-aware)
Low	85	0.776	0.729
Mid	93	0.914	0.903
High	53	0.868	0.830

- High—Low accuracy gap: **0.091 without** quality-awareness vs. **0.101 with** it.
- The intervention **widened**, rather than narrowed, the quality-performance gap.
- Direct negative result — reported as such.

3. Histopathology-Confirmed vs. Panel-Consensus

Group	N	Accuracy
Histopathology-confirmed	28	0.750
Panel-consensus-only	203	0.867

- Model performs *worse* on the histopathology-confirmed subset.
- Plausible reason: histopathology is ordered more often for clinically *ambiguous or suspicious* lesions in real practice — these are inherently harder cases.
- $N = 28$ is too small for a confidence interval — treated as a **qualitative** finding only.

4. Metadata Importance — Does the Model “Look at” the Right Fields?

Method: ablate each field (replace with “unknown”), measure change in output logit.

Highest ranked	Importance
Organ transplant history	0.046
Lesion status when captured	0.041
Referral diagnosis*	0.024
Sex*	0.015
Location group*	0.011
Lowest ranked	
Age / Weight / Diameter* / Height	≤ 0.0011

* = field the dataset paper found significantly associated with malignancy.

Referral diagnosis, sex, location group align with the dataset paper. But **diameter** ranks lowest — likely an architectural artifact (12-d categorical embeddings vs. 1 scalar), not true unimportance.

Comparison with the Literature

Source	Task	N	Bal. acc.
This work (channel-gated)	binary	234 lesions	0.783
This work (SAM + TTA)	binary	234 lesions	0.822
DiffMIC, PAD-UFES-20 (2025)	binary	2298 images	0.836

- Closest verified binary-task comparator, on a similar-shape (image + metadata) dataset.
- We reach within 0.014 of it, with $\sim 10\times$ **less training data**.
- No prior benchmark exists on MCR-SL itself, so this is not framed as “beating state of the art” — the robustness analyses are the paper’s real contribution.

Limitations

- Small dataset: 234 usable lesions, single institution.
- Ground truth mostly panel-consensus (only 28/240 lesions histopathology-confirmed).
- Several 9-class categories have <10 lesions — exploratory only.
- Quality terciles are modest in size ($N \approx 53\text{--}93$) — correlation results are not statistically decisive.
- Metadata-importance comparison is confounded by embedding-dimension imbalance (categorical vs. numeric fields).

Conclusion & Future Work

What we found

- First benchmark on MCR-SL: channel-gated fusion reaches 0.783 balanced accuracy; SAM + TTA pushes this to 0.822.
- Quality-aware training, contrastive loss, and alternative optimizers did **not** help at this scale.
- No significant quality-performance relationship detected; quality-aware training *widened* rather than narrowed the tercile gap.
- Model performs worse on histopathology-confirmed lesions; metadata importance partially aligns with the dataset paper's own statistics.

Future work

Give numeric fields more representational capacity; try text-templated metadata encoding; revisit quality-aware training with a different loss weighting.

Thank you

Questions?